THE HILB FUNCTOR IS AN FPQC SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 281-286 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 288-303, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_pic.json (pb=911, stmt 281-286, proof 288-303).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 03I0: sources/stacks-pic/pinned-original-source.tex, source lines 303–303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02VO: sources/stacks-pic/pinned-original-source.tex, source lines 303–303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 022G: sources/stacks-pic/pinned-original-source.tex, source lines 303–303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 022D: sources/stacks-pic/pinned-original-source.tex, source lines 303–303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes. The functor $\mathrm{Hilb}^d_{X/S}$ satisfies the sheaf property for the fpqc topology (Topologies, Definition 022G).*

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $\{T_i \to T\}_{i \in I}$ be an fpqc covering of schemes over S. Set $X_i = X_{T_i} = X \times_S T_i$. Note that $\{X_i \to X_T\}_{i \in I}$ is an fpqc covering of X_T (Topologies, Lemma 022D) and that $X_{T_i \times_T T_{i'}} = X_i \times_{X_T} X_{i'}$. Suppose that $Z_i \in \mathrm{Hilb}^d_{X/S}(T_i)$ is a collection of elements such that Z_i and $Z_{i'}$ map to the same element of $\mathrm{Hilb}^d_{X/S}(T_i \times_T T_{i'})$. By effective descent for closed immersions (Descent, Lemma 03I0) there is a closed immersion $Z \to X_T$ whose base change by $X_i \to X_T$ is equal to $Z_i \to X_i$. The morphism $Z \to T$ then has the property that its base change to T_i is the morphism $Z_i \to T_i$. Hence $Z \to T$ is finite locally free of degree d by Descent, Lemma 02VO. $\square$